Lecture 5

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convergent sequences.

Monotone conv. thm : mono. + bdd $\rightarrow$ cgt

1) mono. $\uparrow$ + bdd above $\rightarrow$ cgt $\rightarrow$ sup.

2) mono. $\downarrow$ + bdd below $\rightarrow$ cgt $\rightarrow$ inf

note : lim is always unique.

(*) Every cgt seq. is bdd.

$\cup$ LTA need not be true.

$\cancel{\{n^2\}}$ $\quad (-1)^n = \{-1, 1, -1, 1 \ldots\}$

eg. $\{n^2\} = \{1, 4, 9 \ldots\}$

→ div. sequences.

$\{-n^2\}$, $\{n\}$

① mono. ↑ + unbdd above → div $+\infty$

2) mono. ↓ + unbdd below → div $-\infty$

→ oscillating sequences

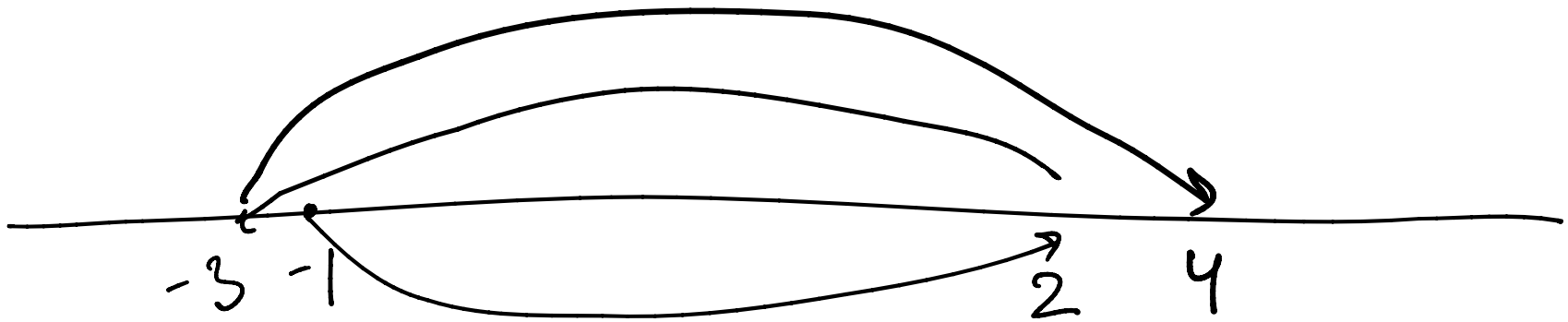
A seq. that neither conv. nor diverges is osc.

1) osc. finitely $\{(-1)^n\}$

2) osc. infinitely

$\{(-1)^n n\}$

$= \{-1, 2, -3, 4, -5, 6 \ldots\}$



1) bdd & not conv $\rightarrow$ osc. finitely

2) Unbdd & not div $\rightarrow$ osc. infinitely

Algebra <u>of sequences</u> $\rightarrow$

$$\left\{ \frac{n^2+3}{2n+1} \right\} \qquad \left\{ \frac{n+3}{2n+1} \right\} \qquad \left\{ \frac{3}{2n+1} \right\}$$

Sandwich thm :

$$a_n \le b_n \le c_n$$

$$a_n \to l, \quad c_n \to l \quad \Longrightarrow \quad b_n \to l$$

Eg: $\left\{ \dfrac{\sin n}{n} \right\}$

$$-1 \le \sin n \le 1$$

$$-\frac{1}{n} \le \frac{\sin n}{n} \le \frac{1}{n}$$

$$-\frac{1}{n} \to 0, \quad \frac{1}{n} \to 0 \quad \Longrightarrow \quad \frac{\sin n}{n} \to 0$$

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